Spotify Engagement from Metadata: Experiment Synthesis

June 12, 2026

Abstract

This document is the living synthesis of the taste-fit prediction campaigns; the markdown reports in `experiments/` are the primary record. The active target is a binary *rotation* signal — did a track earn a repeat listen, play_count ≥ 2 — predicted from a track's *intrinsic metadata* alone (genre, artist, artist/track popularity, follower count, release year, release type) over the user's listening corpus. The headline metric is the area under the ROC curve (AUC); the engagement-magnitude regression target that the lab opened on (play_count $\times$ completion_ratio, $\log 1p$-transformed) is retained as a documented scientific field but is no longer delivered. The behavioral co-listening embedding and all playback/curation fields are deliberately excluded from the feature set. Five Jun-2026 campaigns are folded in below; they converge on a single finding — categorical artist/genre/album identity read by gradient-boosted trees beats every dense modality and every dense distance/margin family — and they crowned the project's first registered, promotable champion, `xgboost-classifier-rotation`, at AUC 0.7429 on the original DS-META corpus. Three Jun-12 campaigns then re-ran the family question on a larger 2026 corpus (a full-64-d, 23 529-row dataset): a four-family classifier bake-off in which `xgboost-classifier` again wins (AUC 0.7204), `catboost` follows in-band, and a dense `pyramid-mlp` is last; a six-geometry architecture scan that closes the dense-net family for AUC by *refuting* the budget hypothesis; and a topology bake-off of a new Rust/`burn` `burn-deep-classifier` that, by feeding the identity one-hots *raw*, repairs the dense net's calibration completely (log-loss 2.0 $\rightarrow$ 0.62) and lifts its AUC $\sim$2 bands to 0.7038 — yet still lands $\sim$1–2 bands short of the tree tier, with learned entity embeddings starved by the low identity cardinality. A seventh campaign then closes that thread: a paired re-run of the topology bake-off on a *higher* identity-cardinality matrix (artist cap 60 $\rightarrow$ 300, genre 40 $\rightarrow$ 120) flips the ranking — `embeddings` now *beats* `mlp` (a +0.0130 swing that reverses the prior result, confirming the starvation hypothesis) — and lifts the net into the in-dataset tree tier (AUC 0.7192, within $\sim$1 band of `catboost`), the first ANN to clear the rotation save bar, saved as `burn-deep-embeddings-highvocab`. Section 6 consolidates why artificial neural nets structurally underperform on this one-hot-heavy feature matrix, how the representation-aware net resolves the diagnosis, and how enough identity resolution finally lets the embedding topology pay off.

1 Introduction

This document is the living synthesis of the experiment campaigns; the markdown reports in `experiments/` are the primary record. Each section transcribes (never recomputes) the quantitative results of a campaign, preserves its run and dataset identifiers verbatim for traceability, and adds the rationale and interpretation that the terse reports omit.

Task and target. The lab predicts *taste fit* from a track's intrinsic metadata over the user's listening corpus. It opened on an engagement-magnitude regression target — play_count $\times$

completion_ratio, trained against a $\log 1p$-transform and mapped back with expm1 — but on 2026-06-08 the active target pivoted to a binary *rotation* signal: play_count ≥ 2, i.e. did the track earn a repeat listen (Section 3). Rotation is the actionable input to the Playlist Pathfinder's per-track desirability term, it is far more learnable than engagement magnitude (trees reach AUC ≈ 0.73 on rotation versus $R^2 \approx 0.3$ on engagement), and with a base rate near 44.5 % it is naturally spread for path search. Engagement is retained as a documented scientific field, but rotation is the delivered target and every leaderboard below is on it.

Metrics. Because the target is binary, the headline metric is the area under the ROC curve (AUC, higher-better), the quantity the classification leaderboard (Table 1) sorts on and the best-models group auto-promotes by; it is reported alongside log-loss, accuracy and the Brier score. Two distinct metric regimes nonetheless coexist in the rotation campaigns. The campaigns opened with a *regression* read — mean absolute error (MAE), RMSE and R^2 on the $[0, 1]$ P(rotation) scale — and only later adopted the honest *classification* read once the framework gained a binary mode. MAE is retained on the historical regression board (Table 4) for same-family band significance, but it is documented there as *degenerate* on a binary target: an ϵ-tube SVR can post a deceptively low MAE while being worse than predicting the mean (Section 5.2), an illusion the classification metrics dissolve. The absolute-percentage metrics (MAPE/medAPE) are undefined on rotation, since half the labels are exactly zero. The historical engagement board, where MAE *is* the natural headline, is preserved in `PROJECT-FACTS.md`; its numbers live on the $\log 1p$ engagement scale and are not comparable to anything below.

Features. The feature space is a track's *intrinsic metadata* alone: genre, artist, artist/track popularity, follower count, release year, credited-artist count and release type. The dense embedding modalities (co-listening word2vec, content-text, acoustic, and the song-autoencoder latent of Section 5.1) are *behavioral or descriptive* signals, not categorical identity, and the payload-based predictors over the metadata one-hots are the canonical models here. Critically, the recipe is *leakage-free*: the playback fields that mechanically constitute or proxy the target — `play_count`, `completion_ratio`, `skip_rate`, `shuffle_rate`, `distinct_sessions`, and the demoted `engagement` field itself — and the behaviorally entangled curation fields are all excluded by default.

Three feature matrices. The first three campaigns ran on the original *DS-META* corpus (12 256 rows, metadata-only). The Jun-12 campaigns ran on a larger 2026 corpus, the *full-64-d* dataset (`ds-20260609-204419-p64-s42`, 23 529 rows, 177 columns: 64 song-AE PCA latent + 7 numeric + 106 identity one-hots at vocab caps artist 60 / genre 40 — the *60/40-vocab* matrix). A final Jun-12 campaign (Section 6.8) re-ran on a *high-vocab* variant (`ds-20260612-143014-p64-s42`, *same* 23 529 rows / split seed 42 / PCA64, differing *only* in identity cardinality: artist cap 60 $\rightarrow$ 300, genre 40 $\rightarrow$ 120; 497 columns including 426 one-hots — artist 301, genre 121, album-type 4). **These are three non-comparable matrices**: DS-META differs from the full-64-d corpus in both size (12 256 versus 23 529 rows) and feature set (metadata-only versus +64 AE latent), and the 60/40 and high-vocab matrices differ in identity cardinality (177 versus 497 columns). AUC numbers must therefore never be read across them — not against the 0.7429 DS-META champion, and not between the 60/40 and high-vocab boards. Each matrix carries its own leaderboard (Tables 1, 2 and 3); nothing regressed when the 60/40 xgboost lands at 0.7204, and every family scores *higher* on the wider high-vocab matrix because the extra identity granularity is genuine signal.

2 Background: model families and metrics

This section gives the minimum theory for a reader who knows ML but not this repo. It is keyed to `PROJECT-FACTS.md` and grows only when a new predictor family or metric enters the reports.

The families in play.

Gradient-boosted trees (`xgboost`, `xgboost-classifier`, `catboost-classifier`, `ngboost-classifier`). An additive ensemble of shallow trees, each fit to the residual gradient of the previous stage. As a *regressor* its output is a leaf average, bounded by the range of training targets; as a *classifier* it emits a calibrated probability through a logistic link, the form that wins on rotation. Trees split on individual columns, so they read a track's sparse artist/genre/album identity one-hots *losslessly* and capture non-linear interactions without manual feature crosses — the reason they dominate this corpus. `catboost` is an ordered-boosting GBDT variant with symmetric (oblivious) trees and native categorical handling; on this corpus it is the legitimate second tree family, co-leader *in-band* with xgboost. `ngboost` is a natural-gradient boosting variant that fits a full predictive *distribution*; its probabilistic machinery, paid for with shallow base learners, costs it raw ranking power here. The cost of the GBDT family generally is a non-deterministic fit (subsampling and column sampling carry an internal RNG seed), so two runs at different seeds differ slightly. The registered champion `xgboost-classifier-rotation` is this family.

Logistic regression. Linear log-odds passed through a sigmoid: a deterministic, ℓ_2-regularized linear classifier. It is the natural *linear floor* for a binary target — the analogue of ridge in the regression world — and serves as a cheap, reproducible control on the harness. Being a single linear boundary in a dense feature space, it blurs the sparse categorical identity that trees exploit, and it lands well below them on rotation.

Kernel SVM / SVC (`svc` with rbf / linear / poly / sigmoid kernels). A maximum-margin classifier that operates on kernel-induced distances between points. As a *regressor* (SVR) on a binary target the ϵ-insensitive tube degenerates — the source of the binary-MAE trap of Section 5.2; as a proper probability *classifier* (SVC) the divergence vanishes but every kernel still loses to the trees, because a dense-distance / large-margin model cannot read sparse identity one-hots the way an axis-aligned tree split does. The sigmoid kernel is the weakest and is the documented refutation echo.

Pyramid-MLP (`pyramid-mlp-classifier`). A feed-forward artificial neural network with a tapering ("pyramid") stack of hidden layers, ReLU activations and an Adam optimizer, fit on the StandardScaler-ed feature matrix. It is the project's only *dense* (every-input-to-every-hidden-unit) learner on rotation. As Section 6 develops at length, a dense net is exactly the regime that one-hot-heavy taste data fights: it must learn dense weights over a mostly-zero input, and it lands last in the bake-off with a blown calibration that an architecture scan proves is structural, not a budget artifact.

Representation-aware net (`burn-deep-classifier`). A full-Rust/`burn` feed-forward classifier built to repair the pyramid-MLP's pathologies. It splits the feature matrix into a *standardized* continuous block (64 PCA + 7 numeric) and a set of *raw* (un-standardized) one-hot identity groups (artist, genre, album-type), so the scaler never turns rare categories into outlier spikes (Section 6.2). A `topology` enum selects among a plain dense `mlp` over the concatenated

inputs, learned per-group `embeddings`, and a `wide_deep` split (a wide linear path over the one-hots beside a deep path over the continuous block). As Section 6.7 reports, the raw-one-hot representation fixes the calibration completely and lifts AUC ∼2 bands over the pyramid, but the net still sits ∼1–2 bands below the tree tier, and the embedding variants gain nothing because the identity cardinalities here are too low to feed them.

Autoencoder-XGBoost blend (`autoencoder-xgboost`). A weighted ensemble $\hat{y} = w \cdot \hat{y}_{\text{ae}} + (1-w) \cdot \hat{y}_{\text{xgb}}$ that combines a regressor over a precomputed 64-dimensional song-autoencoder latent with a metadata-XGBoost model, the weight w tuned on a train holdout. Its purpose was to test whether the dense AE latent carries rotation signal the metadata one-hots miss; it does not, and the tuner correctly drives $w \to 0$, collapsing the ensemble to metadata-only XGBoost (Section 5.1).

Why a binary target makes MAE degenerate. For a continuous target, mean absolute error is a faithful summary of how far predictions sit from the truth. On a $\{0, 1\}$ target it is not: a model can minimize MAE by hugging the labels regardless of how it *ranks* or *calibrates* them. An SVR with an ϵ=0.1 tube snaps its outputs to a two-level step near $\{0.1, 0.9\}$, sitting only ~ 0.1 off every label — a low MAE — while its ranking is worthless (negative R^2). The metrics that matter for a binary taste-fit signal are therefore the classification ones:

AUC (area under the ROC curve). The probability that a random positive is ranked above a random negative; 0.5 is chance, 1.0 is perfect. It is threshold-free and scale-free, depends only on the *ordering* of the predicted probabilities, and is exactly the property the Pathfinder needs from its desirability term. It is the headline metric and the leaderboard sort key.

Log-loss. The negative log-likelihood of the labels under the predicted probabilities; it punishes *confident* mistakes severely and so rewards calibration, not just ranking.

Brier score. The mean squared error between predicted probability and the $\{0, 1\}$ label — a calibration-and-refinement measure on the natural $[0, 1]$ scale, lower-better.

A model that wins on AUC *and* log-loss *and* Brier, as the champion does, is winning on ranking and calibration at once; the binary-MAE artifact that flatters a degenerate SVR shows up on none of them. The same triad later exposes the opposite pathology in a dense net (Section 6): a model can post a respectable AUC — genuine ranking — while its log-loss and Brier are catastrophic, i.e. it ranks acceptably but is *confidently wrong*.

A note on the historical log-target. The lab's opening regression target, engagement, is heavy-tailed and non-negative, so it was trained against a $\log 1p$-transform and mapped back with expm1. That transform pair carries a hazard — a model unbounded in log-space can produce a large positive log prediction that expm1 amplifies into an enormous value, the "expm1 blow-up" — which is why bounded-by-construction families (trees, rbf kernels) are preferred over unbounded ones (linear models, MLPs) on that target. Engagement remains a documented field and its leaderboard lives in `PROJECT-FACTS.md`, but it is no longer the delivered target, so the blow-up hazard is now a historical footnote rather than a live design constraint.

3 The rotation pivot: from engagement magnitude to a binary taste-fit signal

On 2026-06-08 the active target pivoted away from engagement magnitude to a binary *rotation* signal, defined as play_count ≥ 2 — did the track earn a repeat listen? Rotation is the actionable input to the Playlist Pathfinder's `W_FIT` term, where a per-track $[0, 1]$ desirability biases the A* path; a calibrated rotation probability is a cleaner `fit_norm` than a $\log 1p$-then-expm1 engagement estimate. Rotation also turns out to be far more learnable than engagement magnitude (gradient-boosted trees reach AUC ≈ 0.73 on rotation versus $R^2 \approx 0.3$ on engagement) and is naturally spread for path search, with a base rate near 44.5 %. Engagement is retained as a documented scientific field but is no longer the delivered target.

The pivot is decisive for how every number below must be read. Rotation lives on the $[0, 1]$ scale, so its MAE/R^2 are *not* comparable to the engagement leaderboard preserved in `PROJECT-FACTS.md`; and because half the rotation labels are exactly zero, the absolute-percentage metrics (MAPE/medAPE) are undefined for those rows. Two distinct metric regimes therefore coexist in the rotation campaigns: an early *regression* read (MAE, RMSE, R^2, units P(rotation)) and the later, honest *classification* read (AUC, log-loss, accuracy, Brier). As the campaigns below show, the choice of metric is not cosmetic — under MAE a kernel SVR can *look* competitive while being worse than predicting the mean, an illusion that the classification metrics dissolve.

4 Rotation leaderboards

Four leaderboards. The first two are on the original DS-META corpus, one per metric regime; the third and fourth are on the larger 2026 full-64-d corpus. **None of the three AUC boards is comparable to the others** — they sit on three non-comparable feature matrices (DS-META metadata-only; the full-64-d *60/40-vocab* matrix; and the full-64-d *high-vocab* matrix, which shares the 60/40 corpus, rows and split but raises the identity cardinality caps). Each board carries its own in-dataset references and must never be read across (see the Introduction's "Three feature matrices" note). The DS-META AUC board carries the project's first registered, promotable rotation champion; the 60/40 board carries the new-corpus registered xgboost definition; the high-vocab board carries the first ANN to clear the rotation save bar (Section 6.8).

Table 1: Rotation, *classification* regime, **DS-META corpus** — ranked by AUC (higher-better). The champion is the only registered, auto-promoted definition. Datasets: DS-META `ds-20260609-141622-p1-s42` (115 cols, metadata-only, pca floor); the 178-col row adds the 64 song-AE latent columns (`ds-20260608-204402-p64-s42`). AUC noise band $2\sigma = 0.0126$.

Model	Dataset	AUC	log-loss	accuracy	Registered?
xgboost-classifier (champion)	DS-META	**0.7429**	**0.5883**	0.6846	**yes (rank 1)**
xgboost-classifier (178, +AE latent)	p64-s42	0.7350	0.6045	0.6818	no
svc (poly d=3, best non-tree)	DS-META	0.6855	0.6452	0.6450	no
logistic (C=1, linear floor)	DS-META	0.6762	0.6329	0.6491	no
svc (rbf)	DS-META	0.6722	0.6469	0.6426	no
svc (linear)	DS-META	0.6363	0.6535	0.6373	no
svc (sigmoid, worst)	DS-META	0.6255	0.6673	0.6267	no

Table 2: Rotation, *classification* regime, **new full-64-d corpus** — ranked by AUC (higher-better). Dataset `ds-20260609-204419-p64-s42` (23 529 rows, 177 cols: 64 song-AE PCA + 7 numeric + 106 identity one-hots). **Not comparable to Table 1**: different corpus size and feature set; the only valid in-corpus reference is the 0.7204 xgboost point, which the bake-off reproduced byte-identically. The winner's default config is saved as the new-corpus definition. The single-seed sweep best (depth 8, lr 0.03, 0.7261) is +0.0057 over the default — inside the band — so it is flagged for a 3-seed confirm, not promoted. The pyramid-mlp log-loss is its documented calibration ceiling (Section 6); the `burn-deep-classifier` rows are the Rust/`burn` representation-aware net that repairs it (log-loss →∼0.62) yet still sits below the tree tier, so neither is registered (Section 6.7). AUC band $2\sigma \approx 0.0126$ (old-corpus estimate, approximate here).

Model	max_depth/lr	AUC	log-loss	accuracy	Note
xgboost-classifier (winner, saved)	6 / 0.05	**0.7204**	0.6075	0.6585	registered (new
xgboost-classifier (sweep best, single-seed)	8 / 0.03	0.7261	**0.6044**	**0.6651**	flagged for 3-se
catboost-classifier (2nd tree, in-band)	6 / 0.05	0.7111	0.6132	0.6560	−0.0093, inside
burn-deep-classifier (mlp, best tune)	lr 0.0005 / [128,64]	0.7038	0.6221	0.6521	best ANN; cali
burn-deep-classifier (mlp, Phase-1 winner)	lr 0.001 / [128,64]	0.7001	0.6211	0.6470	dense net; calib
ngboost-classifier	—	0.6917	0.6293	0.6392	∼2.3 bands ba
pyramid-mlp-classifier (arch-scan best)	[512,256]	0.6792	2.0584	0.6326	dense net; calib
pyramid-mlp-classifier (bake-off, 300 iter)	[256,128,64]	0.6781	2.4085	0.6326	dense net; calib

5 Campaigns

5.1 Autoencoder-latent + XGBoost blend: a dead modality, isolated

Rationale. This was the first campaign on the pivoted rotation target. The hypothesis was that a precomputed 64-dimensional song-autoencoder latent (from the Qdrant collection `spotify_tracks_song_ae`) carries a track's *identity* in a way the metadata one-hots do not, and that blending an AE-latent regressor with a metadata-XGBoost model — $\hat{y} = w \cdot \hat{y}_{\text{ae}} + (1 - w) \cdot \hat{y}_{\text{xgb}}$, with the weight w tuned on a train holdout — could beat metadata-XGBoost alone. All six arms ran on `ds-20260608-204402-p64-s42` (64 AE latent + 114 metadata = 178 cols, 12,256 rows, 9,805/2,451 at seed 42), varying only the AE branch (kNN at $k \in \{10, 25, 50, 100, 200\}$, plus a ridge-on-latent arm).

Results. All six runs (`run-20260609-131545-e34e4` and siblings) succeeded, and all returned byte-identical blended metrics — MAE 0.38789, R^2 0.191, RMSE 0.4475 — because the weight tuner drove $w \to 0$ in every arm. The only axis that moved was the AE branch's standalone error mae_{ae}, which grew monotonically with the neighborhood size, from 0.4259 at $k = 10$ to 0.4702 at $k = 200$ (the ridge arm landed at 0.4623, between $k = 100$ and $k = 200$). Figure 1 shows the curve against the three horizontal references that matter: even the best AE read ($k = 10$, 0.4259) never reaches the metadata-only tree (0.3879), nor even the AE-latent-only *xgboost* on record (0.4098).

Interpretation. The collapse to $w = 0$ is the correct answer, not a bug to fix: given an AE branch strictly worse than the XGB branch on the holdout, the optimal convex weight *is* zero. The architecture's one real value was that it *isolated* the latent into a branch the tuner could discard, recovering a cleaner result than feeding the latent into a unified XGBoost — which actually *dilutes* the tree (MAE 0.3965 with the 64 latent columns present versus 0.3879 with them dropped). The song-AE latent thus joins co-listening, content-text and acoustic features as a dense modality that loses to categorical identity one-hots. The one caveat at the time was honesty about significance:

Table 3: Rotation, *classification* regime, **high-vocab full-64-d matrix** — ranked by AUC (higher-better). Dataset `ds-20260612-143014-p64-s42` (23 529 rows, 497 cols: 64 song-AE PCA + 9 numeric + 426 identity one-hots — artist 301, genre 121, album-type 4). **Not comparable to Tables 1 or 2**: it shares the 60/40 corpus, rows and split but raises the identity cardinality (artist cap 60 → 300, genre 40 → 120), so every family re-ran here for a fresh in-dataset ranking. The trees moved *up* on the wider matrix (xgboost 0.7204 → 0.7299, catboost 0.7111 → 0.7252). The saved `burn-deep-classifier` (`embeddings`) is the first ANN to reach the in-dataset tree tier and clear the rotation save bar (Section 6.8); xgboost is the in-dataset leader but no tree save rule fired this campaign. AUC band $2\sigma \approx 0.0126$ (old-corpus estimate, approximate here).

Model	config	AUC	log-loss	accuracy	Note
xgboost-classifier (in-dataset leader)	d6 / lr0.05	**0.7299**	**0.6022**	**0.6719**	not registered
catboost-classifier (2nd tree)	defaults	0.7252	0.6068	0.6649	−0.0047, in ba
burn-deep-classifier (embeddings, Phase-2 best)	ed8 / lr0.0005	0.7228	0.6105	0.6645	tree tier; sub-b
burn-deep-classifier (embeddings, *saved*)	ed16 / lr0.001	0.7192	0.6087	0.6694	registered (hig
burn-deep-classifier (mlp)	lr0.001	0.7129	0.6166	0.6617	−0.0063 vs em
burn-deep-classifier (wide_deep)	lr0.001	0.7061	0.6256	0.6526	laggard, both

Table 4: Rotation, *regression* regime, **DS-META corpus** — ranked by MAE on the $[0, 1]$ P(rotation) scale (lower-better). The metadata-only XGBoost leader is the `mae_xgb` branch reproduced inside every blend run; it was never saved as a standalone definition. The SVM-poly row is listed for completeness and *refuted*: its lower MAE is the ϵ-tube artifact of Section 5.2, its R^2 is negative. MAE noise band $2\sigma = 0.0086$.

Model	MAE	RMSE	R^2	Source
metadata-only XGBoost (AE latent dropped)	**0.3879**	**0.4475**	**0.191**	ae-blend / svm
xgboost on latent+metadata combined	0.3965	—	0.166	rotation pivot
xgboost on AE-latent only (64 cols)	0.4098	—	0.132	rotation pivot
svm (poly, ϵ-tube artifact — REFUTED)	0.3814	0.5303	−0.137	svm

the 0.3879 versus 0.3965 edge is only 0.0086 MAE on a $[0, 1]$ target, and the rotation noise band was then unmeasured — so the campaign reported the number but asserted no significant win and promoted nothing. *Source: `2026-06-09-ae-xgboost-blend.md`.*

5.2 Kernel SVM on rotation: a refutation, and the binary-MAE trap

Rationale. With the latent ruled out, this campaign tested whether any SVM kernel (rbf / linear / poly / sigmoid) produces a rotation predictor competitive with the metadata-only XGBoost leader, and — only if one passed — whether it added orthogonal signal in a grid-weighted blend. The metadata matrix was reduced to the pca floor in a new dataset (`ds-20260609-141622-p1-s42`, 115 cols), with two split-seed siblings (seeds 17 and 101) built to measure the noise band, and the champion reproduced as a control (`run-20260609-141717-9a8ba-autoencoder-xgboost`, blend weight 0). A secondary outcome was the campaign's central deliverable: the **first measured rotation noise band**.

Results. Across the three split seeds of the metadata-only XGBoost control (s42 0.387891, s17 0.393072, s101 0.396417) the sample standard deviation is $\sigma_{\text{split}} = 0.00430$, giving a 2σ significance

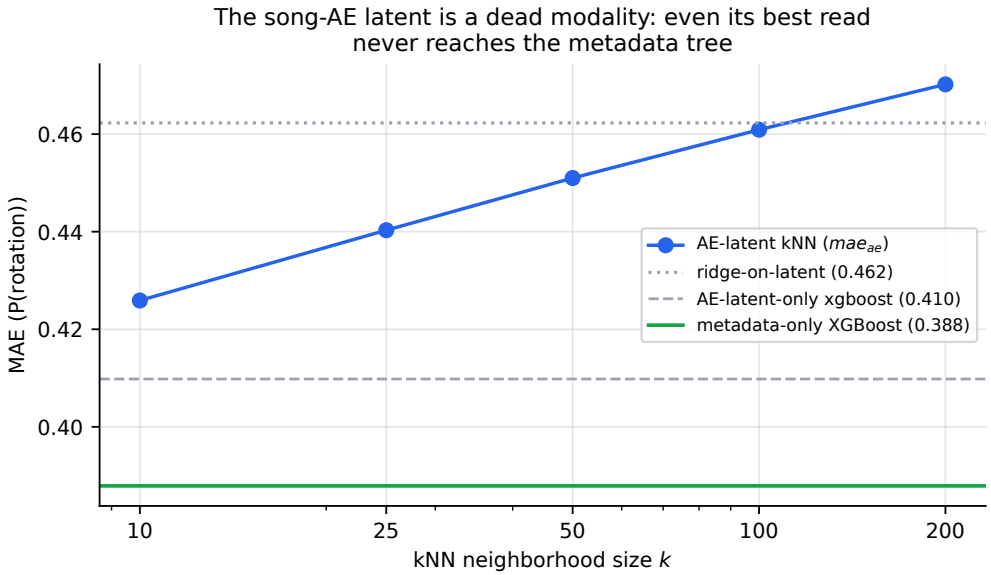


Figure 1: The song-AE latent carries no rotation signal the metadata one-hots do not already carry better. The kNN standalone error mae_{ae} rises monotonically as the neighborhood widens (the latent's signal is strongest in tight local neighborhoods and washes toward the base rate), and even its best point sits above both the AE-latent-only tree and the metadata-only tree. The weight tuner correctly drives the blend weight to zero, so the ensemble collapses to metadata-only XGBoost.

band of **0.0086 MAE** on the $[0, 1]$ scale. This retroactively settles the previous campaign's open question: the 0.3879 versus 0.3965 edge is $\Delta 0.0086 \approx 1\cdot$band (likely better, not confident), while the 0.3879 versus 0.4098 gap (AE-latent-only) is $\sim 2.5\cdot$band and therefore real. As for the SVM arms, rbf, linear and poly all posted MAE inside the control band (0.381–0.390), *lower* than the champion in the poly case (0.3814) — yet every one of them is worse-than-mean on R^2 (-0.13 to -0.20), with RMSE 0.53–0.54 against the tree's 0.4475. Sigmoid SVR diverged outright: MAE 0.624, RMSE 1.531, R^2 -8.48, with predictions reaching 24.7 on a $[0, 1]$ target — the classic unbounded sigmoid-SVR blow-up. Figure 2 contrasts the two metric views directly.

Interpretation. The low SVR MAE is a degeneracy of the metric on a binary target. With an ϵ-tube of width 0.1 on $\{0, 1\}$ labels, the SVR collapses to a two-level step function pinned near the right classes — each row is about 0.1 off its label, so the MAE is small, while calibration and ranking are destroyed (negative R^2). The metric that matters for the Pathfinder `fit_norm` (ranking and calibration, i.e. R^2/RMSE/AUC) says the SVM loses decisively. Worse, the two stage-2 blends (`run-20260609-143114-528ef-blend` and its lgbm twin) fitted weights of exactly $[0.0, 1.0]$ — the grid, optimizing the same degenerate validation MAE, *mis-selected pure SVM* and discarded the tree, the opposite of the self-disabling seen in the blend campaign. The lesson is sharp: a grid-by-MAE blend objective is itself unsafe on a binary target. Nothing was registered or promoted; kernel SVM became the fifth dense-regime family to lose to one-hot identity on this corpus. *Source: `2026-06-09-svm-kernel-rotation.md`.*

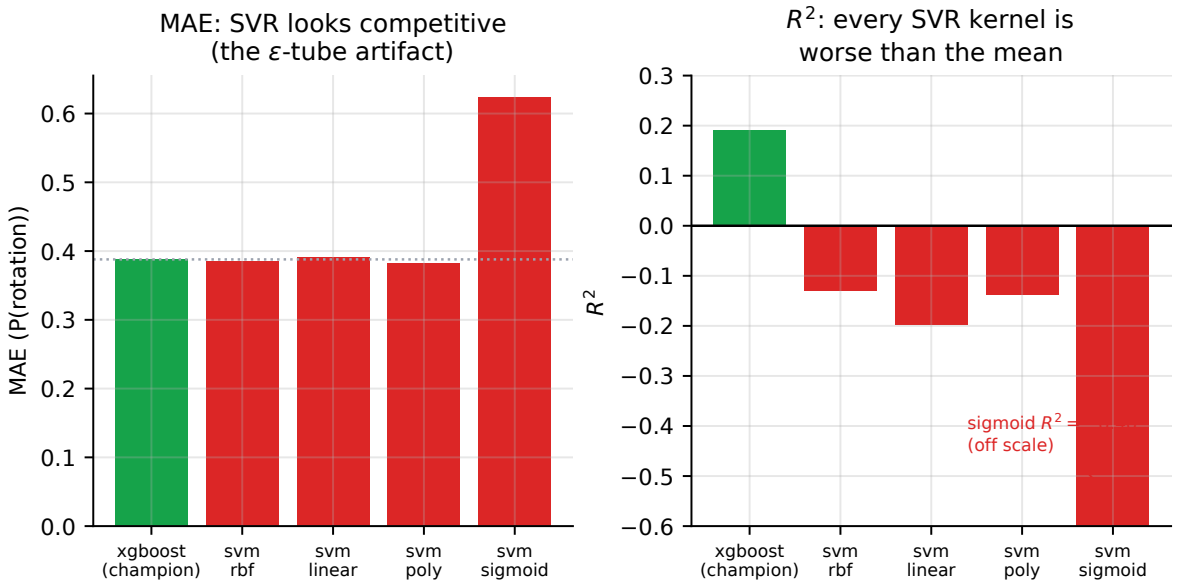


Figure 2: On a binary target, low SVR MAE is an artifact, not skill. With $\epsilon = 0.1$ on a $\{0, 1\}$ target the SVR snaps its outputs to the ϵ-tube edges (a hard two-level step, $p_{50} \approx 0.10$, $p_{90} \approx 0.90$), so each row sits only ~ 0.1 off its label and the *MAE* (left) reads competitive. The honest R^2 (right) exposes it: every kernel is worse than predicting the mean. Sigmoid is off-scale at $R^2 = -8.48$.

5.3 The honest AUC bake-off and the first AUC champion

Rationale. The SVM campaign exposed that MAE is the wrong lens for a binary target. The framework had meanwhile gained a classification mode (`domain.toml task = binary`, primary metric AUC, higher-better; the best-models group ranks by AUC and auto-promotes the top run). This campaign re-ran the comparison on that honest footing: xgboost-classifier against logistic and a four-kernel SVC scan, all on DS-META, plus a 178-column arm with the AE latent re-added to settle the latent-dilution question on the AUC axis, and three control seeds to measure the AUC noise band. The hypothesis — a sixth test of the convergent finding — was that xgboost-classifier wins and every dense family loses.

Results. All nine runs succeeded. The champion config (`run-20260609-172920-627d6-xgboost-classifier`, control hyperparameters, seed 42) reached **AUC 0.7429** (log-loss 0.5883, accuracy 0.6846, Brier 0.2017) and was registered as `xgboost-classifier-rotation` and auto-promoted to best-models rank 1 — the project's first saved, AUC-scored, promotable rotation champion. The first measured *AUC* noise band comes from the three control seeds (s42 0.742934, s17 0.730659, s101 0.734469, mean 0.736020): $\sigma_{\text{AUC}} = 0.00628$, $2\sigma = \mathbf{0.0126}$ (Figure 4). This band is a different metric from the 0.0086 MAE band and does not replace it. Every dense family lost by many bands — svc-poly 0.6855, logistic 0.6762, svc-rbf 0.6722, svc-linear 0.6363, svc-sigmoid 0.6255 — a spread of 4.6 to 9.3 noise bands below the champion (Figure 3). The 178-column arm (AE latent re-added) scored 0.7350, −0.0079 below the metadata-only champion — the same dilutive direction as the regression result, but inside the AUC band, so neutral-leaning-dilutive at a single seed.

Interpretation. Reframing rotation as classification did two things at once. It dissolved the SVR illusion of Section 5.2 — under AUC, log-loss *and* Brier, every SVC kernel is now visibly worse than the trees, all in the same direction, so the ϵ-tube artifact that made SVR's MAE look competitive

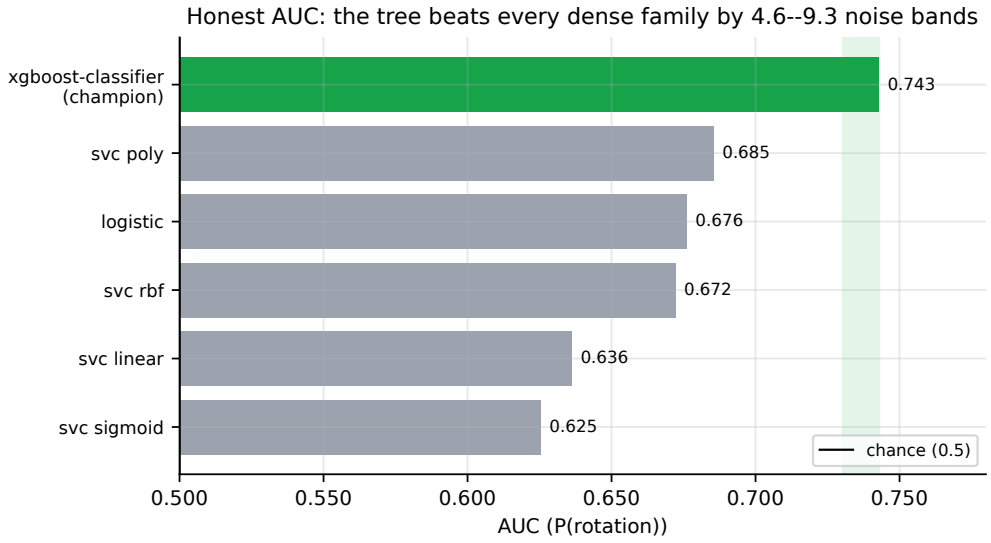


Figure 3: The honest classification metric confirms the convergent finding a sixth time. xgboost-classifier reads the sparse artist/genre/album identity one-hots losslessly; the dense distance/margin families (logistic and all four SVC kernels) blur that identity in a dense feature space and land 4.6 to 9.3 noise bands lower. The shaded strip is the $2\sigma = 0.0126$ band below the champion; no challenger comes within a band of it.

simply vanishes. And it confirmed that the metric was only ever a problem for the *fairness of the comparison*; the dense family was always the loser. Notably svc-sigmoid does not collapse to chance under classification (it holds AUC 0.6255 rather than blowing up): a proper probability objective removes the unbounded divergence that wrecked the SVR-sigmoid run. The convergent finding now stands on an honest metric: the taste-fit signal lives in artist/genre/album *identity*, captured by sparse one-hots and exploited by trees, not in dense distance or margin models. The remaining lever is identity *resolution* — raising the vocabulary caps so the `artist=__other__` tail gets real columns — not new modalities or families. The next planned campaign is a champion hyperparameter scan now that the family and metric are settled. *Source:* `2026-06-09-rotation-classification.md`.

5.4 Four-family bake-off on the new full-64-d corpus, and an xgboost depth/lr sweep

Rationale. The first three campaigns ran on the original DS-META corpus. This campaign re-asked the family question on the larger 2026 corpus — "all of the 64-d embedding", the full 177-column matrix (`ds-20260609-204419-p64-s42`, 23 529 rows, 64 song-AE PCA + 7 numeric + 106 identity one-hots) — by putting four classifier families head to head at registry defaults, seed 42: `xgboost-classifier`, `catboost-classifier`, `ngboost-classifier` and a dense `pyramid-mlp-classifier`. The latter three are *new families on this project*. The winner then earned a `max_depth`× `learning_rate` sweep. The only valid in-corpus reference is a pre-existing run on this dataset (`run-20260609-204726-439d8-xgboost-classifier`, AUC 0.7204); nothing here may be read against the DS-META 0.7429 champion (different corpus and feature set).

Results. All four bake-off arms succeeded (Table 5; Figure 5). `xgboost-classifier` (`run-20260612-115909-840c0-xgboost-classifier`) won at AUC 0.7204, reproducing the in-corpus reference byte-identically (0.720 438 340 3) — the determinism gate passed. `catboost-classifier` (`run-`

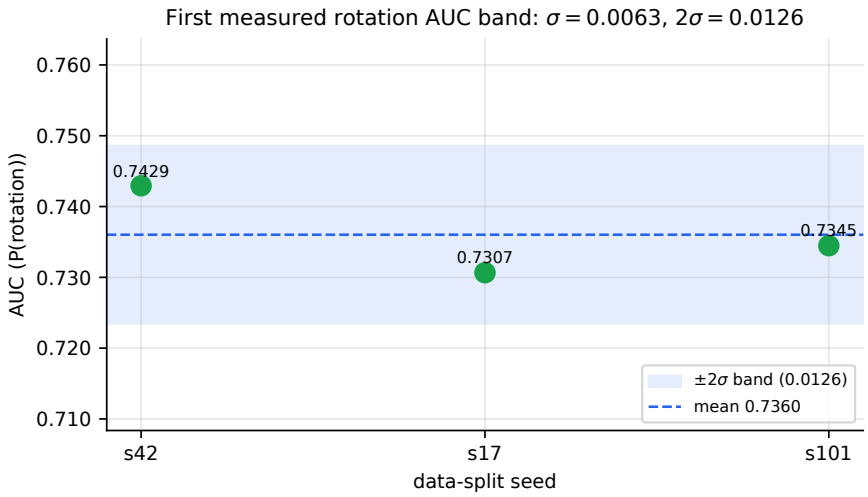


Figure 4: The first measured rotation AUC noise band. Three data-split seeds of the registered champion config give $\sigma_{\text{AUC}} = 0.0063$ and a 2σ significance band of 0.0126. In relative terms this is actually tighter than the MAE band (0.85% versus 1.10% coefficient of variation), as expected for a rank metric on 2,451 test rows. The band only matters for the sub-band latent-dilution call; champion and challengers are separated by many bands.

`20260612-115909-c7ef6-catboost-classifier`) followed at 0.7111, −0.0093 behind — *inside* the ∼0.0126 band, so co-leader and the legitimate second tree family, not a loss. **`ngboost-classifier`** (`run-20260612-115909-d725b-ngboost-classifier`) landed at 0.6917 (∼2.3 bands back), and the dense **`pyramid-mlp-classifier`** (`run-20260612-115909-ea698-pyramid-mlp-classifier`) was last at 0.6781 — with a log-loss of 2.4085 and Brier 0.3251 far above the trees' ∼0.60–0.63 / ∼0.21, the diagnostic dissected in Section 6.

Table 5: Phase 1 — the four-family bake-off on the new full-64-d corpus (`ds-20260609-204419-p64-s42`, seed 42, registry defaults). All four succeeded. AUC higher-better; log-loss and Brier lower-better. Winner bolded; best cell per metric bolded.

Arm / run id	Predictor	AUC	log-loss	accurac
C0 (`run-20260612-115909-840c0-xgboost-classifier`)	**xgboost-classifier**	**0.7204**	**0.6075**	**0.658**
A1 (`run-20260612-115909-c7ef6-catboost-classifier`)	catboost-classifier	0.7111	0.6132	0.656
A2 (`run-20260612-115909-d725b-ngboost-classifier`)	ngboost-classifier	0.6917	0.6293	0.639
A3 (`run-20260612-115909-ea698-pyramid-mlp-classifier`)	pyramid-mlp-classifier	0.6781	2.4085	0.632

The winner earned a Phase-2 sweep over `max_depth` $\in \{4, 6, 8\} \times$ `learning_rate` $\in \{0.03, 0.05, 0.1\}$ (Table-form data in the source report; Figure 6). Depth is the dominant axis: every depth-4 cell is at the bottom (0.7074–0.7119), and the depth-8 / lr-0.03 corner tops the grid at 0.7261 (`run-20260612-120207-2dd3d-xgboost-classifier`, log-loss 0.6044). High learning rate (0.1) consistently hurts both AUC and calibration. But the best-vs-default gap (+0.0057) is *inside* the band, so the default depth-6/lr-0.05 config was saved as the new-corpus definition and the depth-8/lr-0.03 cell was flagged for a 3-seed confirmation rather than promoted.

Interpretation. The convergent finding survives the corpus change intact: on a different, larger corpus with the AE latent present, the two tree families again lead and the dense net is again last.

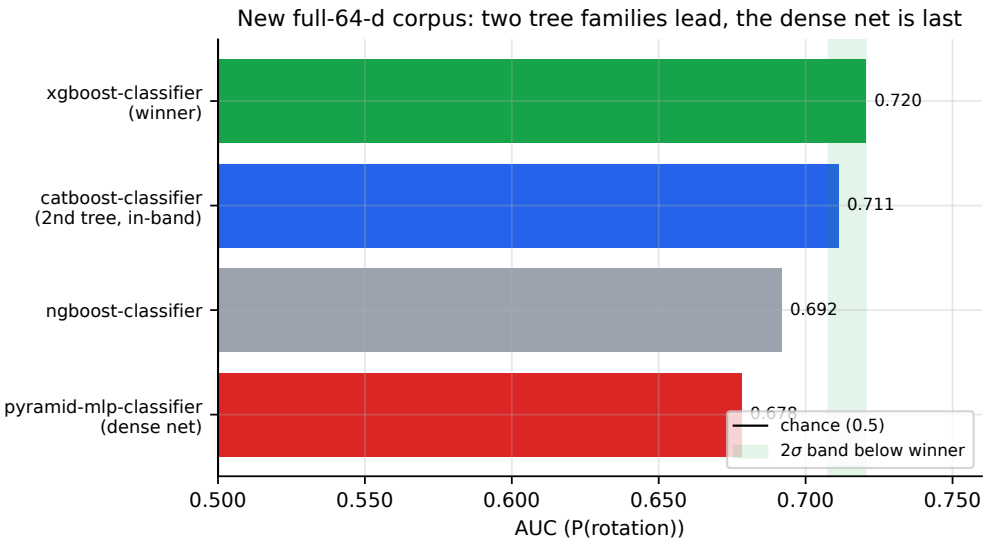


Figure 5: The family ordering survives the corpus change. On the larger 2026 full-64-d matrix the two gradient-boosted-tree families lead (`xgboost` winning, `catboost` co-leader inside the shaded 2σ band below the winner); `ngboost` sits ∼2.3 bands back; the dense `pyramid-mlp` is last, the same convergent finding seen on DS-META in Figure 3.

`catboost` earns its place as a legitimate second tree family (co-leader in-band) — a tree-diversity / second-opinion member, not an AUC gain over xgboost. `ngboost` pays for its distributional machinery: shallow base learners (`base_max_depth` 3) under-fit the 177-column one-hot-heavy matrix. The pyramid-MLP's last-place AUC *plus* its blown log-loss are the dense-net signature; the bake-off read the log-loss as a 300-iteration under-training artifact and prescribed more budget — a hypothesis the next campaign tests and refutes. *Source:* *`2026-06-12-classifier-bakeoff-p64.md`.*

5.5 Pyramid-MLP architecture scan: refuting the budget hypothesis

Rationale. The bake-off flagged the pyramid-MLP's last place (AUC 0.6781, log-loss 2.4085) as a 300-iteration-cap under-training artifact, prescribing "raise `max_iter` so the net converges." This follow-up tests that mechanism directly: six pyramid geometries, all with `max_iter` raised to **800**, alpha 1×10^{-4}, learning rate 1×10^{-3}, seed 42, on the same dataset — so the comparison reflects *architecture*, not budget. Does any geometry, given the budget to converge, lift the dense family to a competitive AUC?

Results. Refuted on every count (Table 6; Figures 7 and 8). **No arm hit the 800 cap**: every geometry self-stopped at **73–100 iterations** via sklearn's `n_iter_no_change` loss-plateau tolerance — far below even the old 300 cap, so the budget was never the binding constraint. The decisive tell: the re-baseline arm M0 ([256,128,64], the bake-off geometry now at `max_iter` 800) reproduced the under-trained run's AUC to all digits — 0.678 092 6 versus 0.678 092 6 — because it ran the identical 73 iterations. Raising the cap from 300 to 800 changed nothing. The best converged geometry was M1 [512,256] at AUC 0.6792 (`run-20260612-123155-0ace5`), a statistical tie with the under-trained default (+0.0011 ≈ zero) and ∼3.3 bands below xgboost. The whole family lands in a 0.0080-wide AUC strip (0.6712–0.6792), entirely inside one noise band — the geometry ranking is not resolvable at a single seed — with the log-loss (2.06–2.78) and Brier (0.32–0.34) blown at the same level as the under-trained default across every arm. No definition was

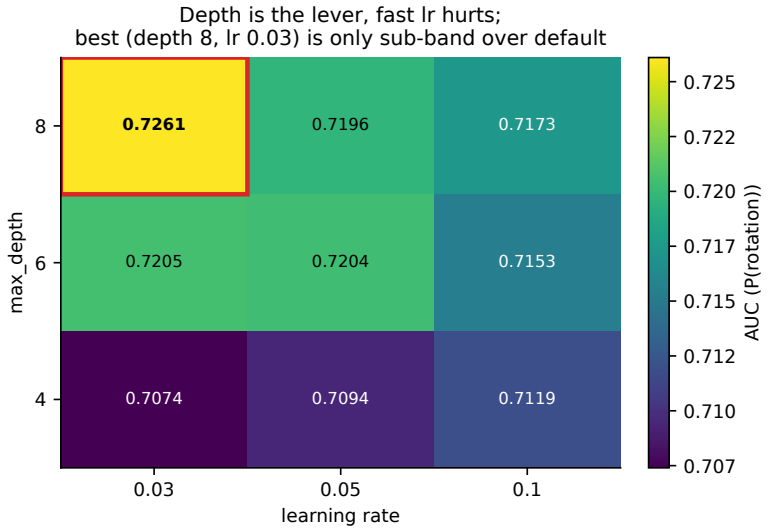


Figure 6: The xgboost `max_depth`×`learning_rate` sweep on the new corpus. Depth is the lever (the AUC climbs monotonically up the depth axis at the slow learning rates), and a fast learning rate hurts. The best cell (depth 8, lr 0.03, boxed) reaches 0.7261 but only +0.0057 over the default — inside the 2σ band — so the default config remains saved and the best cell awaits a 3-seed confirm.

saved (below the 0.683 save bar).

Table 6: The six-geometry pyramid-MLP architecture scan (`ds-20260609-204419-p64-s42`, `max_iter` 800, alpha 1×10^{-4}, lr 1×10^{-3}, seed 42), sorted by AUC. All six succeeded; *none* hit the 800-iteration cap — each self-stopped at the listed iteration count. Best cell per metric bolded.

Arm	Hidden layers	AUC	log-loss	Brier	Iters (of 800)
M1 wide-shallow	[512, 256]	**0.6792**	**2.0584**	**0.3216**	75
M0 re-baseline	[256, 128, 64]	0.6781	2.4085	0.3251	73
M4 gentle taper	[384, 307, 246]	0.6777	2.7764	0.3292	87
M2 deep pyramid	[256, 128, 64, 32]	0.6737	2.7631	0.3303	100
M3 steep taper	[512, 169, 56]	0.6736	2.7031	0.3365	89
M5 wide-deep	[512, 307, 184, 111]	0.6712	2.6457	0.3348	84

Interpretation. The refutation is clean and twofold: neither the training budget (every arm self-stops far below the cap; M0 reproduces the under-trained AUC to all digits) nor the architecture (all six geometries inside one band) moves the dense family. The high log-loss is therefore not a truncated training trajectory but a genuine *calibration ceiling* at these hyperparameters — the optimizer's chosen plateau. With the dense family's own knobs exhausted, the binding constraint must lie elsewhere: in the input *representation*, the subject of Section 6. The directional within-family trend — wider-and-shallower mildly favored (M1 top), depth/capacity mildly harmful (M5 wide-deep last) — is consistent with a dense net struggling to route signal through 106 sparse one-hots: extra layers add capacity to memorize and blur, not to separate. The family is closed for AUC on this feature set; do not re-scan geometry or budget. *Source: `2026-06-12-pyramid-mlp-arch-scan.md`.*

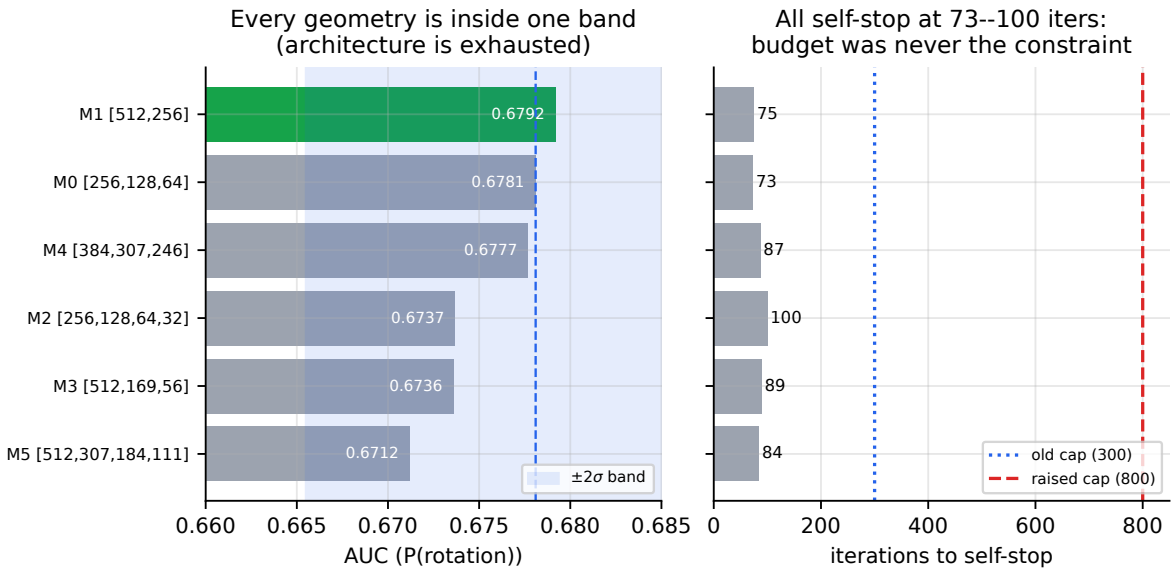


Figure 7: Both knobs of the dense family are exhausted. *Left:* the six geometries land inside a single 2σ band around the under-trained default — architecture is not the binding constraint. *Right:* every geometry self-stops at 73–100 iterations, well below even the old 300 cap (let alone the raised 800), so the training *budget* was never the binding constraint either. The optimizer reaches a loss plateau it cannot escape at this learning rate and halts. The binding constraint is the input *representation*.

6 Why artificial neural nets underperform on one-hot-heavy taste data

The dense `pyramid-mlp` is the only artificial neural net in the rotation campaigns, and across two Jun-12 campaigns it lands last with a distinctive failure pattern. This section consolidates, rigorously, *why* — grounding every claim in the campaign evidence. The short version is the project's documented *convergent finding*: on this feature matrix the binding constraint is the input *representation*, and a dense net is the one model class structurally mismatched to it. We build the argument in five parts and close with the motivated next step.

6.1 The feature matrix is mostly sparse identity

The full-64-d matrix has 177 columns, and their *composition* is the whole story. **106 of the 177 columns are sparse identity one-hots**: artist (×61), `genre_primary` (×41) and `album_type` (×4). Only 71 columns are continuous — the 64 song-AE PCA latents and 7 numeric metadata fields (popularities, follower count, release year, credited-artist count). For any given track exactly one artist column, one genre column and one album-type column is hot; the rest of the 106 are zero. The matrix is, by construction, dominated by a mostly-zero block of mutually exclusive indicators.

A gradient-boosted tree reads this block *losslessly and cheaply*: a single axis-aligned split on column `artist=X` partitions the data on exactly that identity, with no interaction among the one-hots forced and no cost paid for the columns that are zero. Capturing "tracks by artist X in genre Y rotate" costs the tree two nested splits. A dense net cannot do this. It must learn a

dense weight from every one of the 106 mostly-zero inputs into every hidden unit, then back out a non-linear combination — spending capacity and gradient signal modeling the structure of a block that the tree handles with one comparison. This is precisely the regime in which trees structurally dominate dense nets, and it is the project's *convergent finding*, now observed across four feature modalities and five model families: the taste-fit signal is carried by artist/genre identity, captured by one-hots and exploited by trees, not by descriptive attributes or distance/margin/dense models. The pyramid-MLP is simply the dense family's turn to lose to it.

6.2 Standardization turns rare categories into outlier spikes

The dense net is fed a StandardScaler-ed matrix, and standardization interacts pathologically with sparse one-hots. A one-hot column for a category present in a fraction p of rows has mean p and standard deviation $\sqrt{p(1-p)}$. Standardizing it maps the "on" rows to

$$z_{\text{on}} = \frac{1-p}{\sqrt{p(1-p)}} = +\sqrt{\frac{1-p}{p}}, \qquad z_{\text{off}} = \frac{0-p}{\sqrt{p(1-p)}} = -\sqrt{\frac{p}{1-p}}.$$

For a *rare* category — the long tail of artists and genres, where p is small — the "on" value $+\sqrt{(1-p)/p}$ becomes a large positive spike (for $p = 0.01$ it is $\approx +9.9$), while the "off" value sits at a tiny negative $-\sqrt{p/(1-p)} \approx -0.1$. Every rare identity therefore arrives at the net as an occasional, enormous outlier on its own axis. ReLU units with Adam at learning rate 1×10^{-3} swing hard on these spikes — a single rare-category activation drives a large pre-activation that the optimizer chases — and the softmax/sigmoid head saturates, emitting near-0 or near-1 probabilities on thin evidence. Trees are immune: a split threshold on a one-hot is invariant to any monotone rescaling, so the scaler's spike does nothing to them. This is a direct mechanistic contributor to the blown calibration documented next.

6.3 The diagnostic signature: genuine ranking, broken calibration

The dense net's failure has a precise, reproducible signature in the metrics — and it is *not* the failure of a weak model. The pyramid-MLP posts AUC ≈ 0.68 (0.6781 in the bake-off, 0.6792 best in the arch scan): genuine ranking signal, well above the 0.5 chance line, only $\sim$3 bands below the trees. Yet *at the same time* its log-loss is $\approx$ 2.0–2.8 and its Brier $\approx$ 0.32–0.34, against the trees' log-loss $\approx$ 0.60–0.63 and Brier $\approx$ 0.21 (Figure 8). A constant base-rate predictor (always outputting $p \approx 0.445$) scores a log-loss of only ≈ 0.69. **The net's log-loss of 2.0–2.8 is thus far worse than the random baseline, while xgboost's 0.61 is well below it.**

This combination — acceptable ranking, worse-than-random log-loss — means the net is *confidently wrong*: it pushes many predictions to the saturated extremes (near 0 or near 1) on the wrong side of the label, and log-loss punishes each such confident error severely. It is miscalibrated, not weak: the ordering is there, but the probabilities attached to that ordering are unusable as a `fit_norm`. The mechanism is exactly the scaler-spike saturation of Section 6.2. Contrast the trees, whose leaf-frequency outputs are calibrated by construction: a leaf reporting probability q is the empirical rotation frequency of the training rows that fall in it, so the tree's probabilities are honest by design — the reason it wins on log-loss and Brier as well as AUC.

6.4 Optimization versus representation: the binding constraint

A natural objection is that the net is merely under-optimized — too few iterations, the wrong geometry. The architecture scan (Section 5.5) forecloses both. On the *budget* axis, every one of

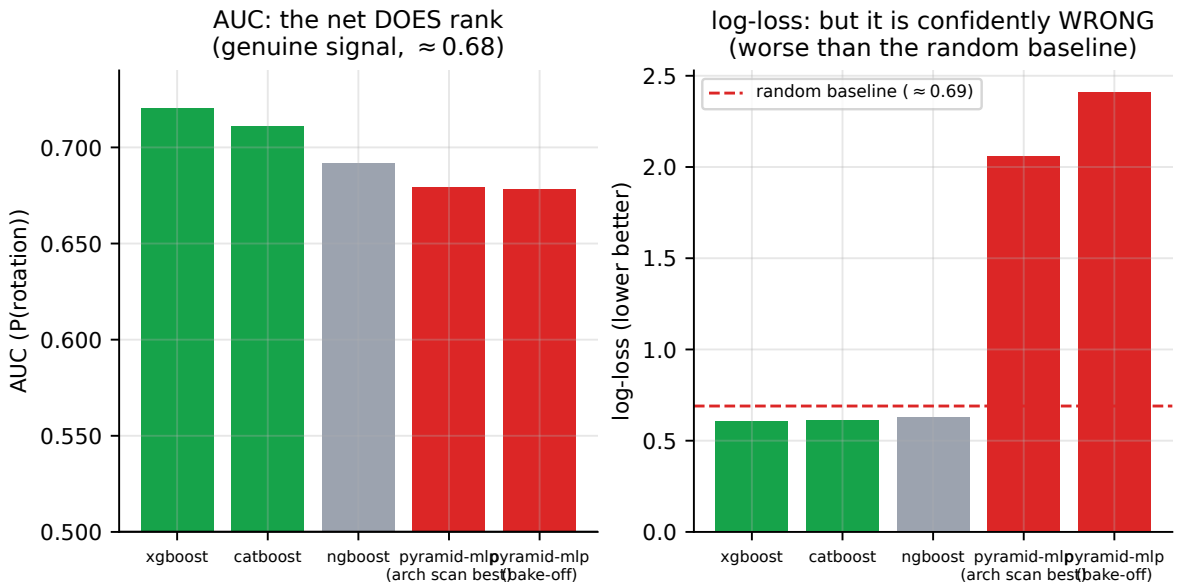


Figure 8: The dense-net diagnostic: the net is confidently wrong, not merely weak. *Left:* on AUC the pyramid-MLP genuinely ranks (≈ 0.68, well above chance, ∼3 bands below the trees) — there is real signal in its ordering. *Right:* but its log-loss (≈ 2.0–2.8) is far *worse* than the random base-rate baseline (≈ 0.69, dashed), while the trees sit well below it (≈ 0.61). A model that ranks acceptably yet scores worse-than-random log-loss is emitting confident, miscalibrated probabilities.

the six geometries self-stops at 73–100 iterations, well below even the old 300 cap (let alone the raised 800): the optimizer reaches a loss plateau it cannot escape, so more budget buys nothing — the re-baseline arm M0 reproduced the under-trained AUC to all digits because it ran the identical 73 iterations. On the *architecture* axis, all six geometries land inside a single noise band, and the best converged geometry reproduces the under-trained AUC to four digits (0.6792 versus 0.6781). Both knobs internal to the dense family are therefore exhausted. By elimination, the binding constraint is neither budget nor architecture but the input *representation* — the raw one-hots of Section 6.1, which the net cannot consume the way a tree does. This is the diagnostic that points at representation-aware topologies (entity embeddings, wide-and-deep) as the only untested lever, since those are designed precisely to replace raw one-hots with a learned dense representation.

6.5 PCA has already pre-empted the net's nonlinear job

There is a second reason the dense net has little to do. The 64 song-AE PCA latents are already a *linear, decorrelated* compression of a 1536-dimensional embedding: PCA has extracted the dominant linear structure and handed the net orthogonal components with most of the easy linear variance already separated. Whatever nonlinear interaction a hidden layer might discover in that block has largely been pre-empted; little nonlinear structure remains for the net's depth to exploit. The evidence is direct: a plain logistic regression reaches AUC 0.676 on this corpus and the converged MLP reaches 0.679 — a dead heat. A multi-layer ReLU network buys essentially nothing over a single linear boundary, which is exactly what one expects when the continuous block is already linearly pre-conditioned and the categorical block is the part that carries signal the net cannot read.

6.6 The open question and the motivated next step

The diagnosis is now sharp: the dense net's ceiling is set by its inability to consume raw one-hot identity, not by budget, architecture, or the continuous block. That isolates a single, testable open question — *can a representation-aware topology close the gap to the trees?* The two ingredients that target the diagnosis directly are (i) **entity embeddings** for the identity groups (artist, genre, album-type), which replace each sparse one-hot block with a learned low-dimensional dense vector — the standard fix for high-cardinality categoricals in nets, removing both the dense-weights-over-zeros problem of Section 6.1 and the scaler-spike problem of Section 6.2; and (ii) a **wide-and-deep** split — a wide linear path over the one-hots (so identity is read directly, as the tree reads it) alongside a deep path over the continuous block. A Rust/`burn` implementation of such a representation-aware network — the `burn-deep-classifier` — was built precisely to answer this question. Section 6.7 reports its results, and they resolve the diagnosis cleanly: the calibration pathology was indeed a representation artifact and is fixed, but the AUC gap to the trees only *narrows*, it does not close.

Sources: 2026-06-12-classifier-bakeoff-p64.md, 2026-06-12-pyramid-mlp-arch-scan.md.

6.7 Resolved: the burn-deep-classifier fixes calibration but not the AUC gap

Rationale. The preceding subsections left a single sharp open question: the dense net's ceiling is set by its inability to consume raw one-hot identity, so *can a representation-aware topology close the gap to the trees?* The `burn-deep-classifier` — a full-Rust/`burn` network — was built to answer it. Its design targets the two diagnosed pathologies directly: it keeps the identity one-hots *raw* (standardizing only the 64 PCA + 7 numeric continuous block), so the scaler-spike mechanism of Section 6.2 never fires, and it exposes a `topology` enum over a plain dense `mlp`, learned entity `embeddings`, and a `wide_deep` split. Phase 1 compared the three topologies (epochs 250, patience 20, early-stop on validation log-loss, hidden $[128, 64]$, embed_dim 8, lr 1×10^{-3}, batch 256); Phase 2 tuned the winner. All nine runs ran on the same full-64-d dataset (`ds-20260609-204419-p64-s42`) and succeeded cleanly.

Results. The headline is the calibration recovery (Figure 9). Every topology lands at log-loss 0.621–0.628 and Brier 0.217–0.220 — squarely in the trees' ∼0.60–0.63 / ∼0.21 regime — against the sklearn pyramid's log-loss 2.0–2.4 and Brier 0.33. **The calibration pathology is fixed completely.** On the topology axis (Table 7), the plain `mlp` wins at AUC 0.7001, with `embeddings` (0.6934) and `wide_deep` (0.6918) both inside the $2\sigma = 0.0126$ band of it and of each other — `embeddings` edges `wide_deep` by a sub-band +0.0016. The Phase-2 `mlp` tune (Table 8) tops out at AUC 0.7038 (lr 5×10^{-4}, $[128, 64]$): the whole 6-cell grid is a 0.0081-wide strip (0.6957–0.7038), every $[128, 64]$ cell beats every $[256, 128]$ cell, and the best config is only +0.0037 (inside the band) over the Phase-1 winner, so the Phase-1 `mlp` config stands. At 0.7038 the net sits −0.0073 below `catboost` (∼0.6 band) and −0.0166 below `xgboost` (∼1.3 bands); below the ∼0.71 save bar, so no definition was persisted.

Interpretation. The result resolves Section 6.6's open question in two parts, and the two answers point in opposite directions. First, the *calibration* verdict is overturned: the pyramid-MLP's worse-than-random log-loss of Section 6.3 was a *representation* artifact — raw one-hots crushed into the standardizer — not an intrinsic property of dense nets on taste data, and a net that keeps the one-hots raw is calibrated in the trees' regime. The earlier "broken calibration is a fixed ceiling" reading holds only for the sklearn-pyramid representation. AUC also recovered: the burn net's 0.7038 is +0.025 over the best converged pyramid (0.6792), ∼2 bands of genuine ranking-power gain, so the prior conclusion that "the MLP family is closed for AUC at ≈ 0.68" is

Table 7: Phase 1 — the `burn-deep-classifier` topology bake-off (`ds-20260609-204419-p64-s42`, held params: epochs 250, patience 20, hidden [128, 64], embed_dim 8, dropout 0.1, lr 1×10^{-3}, batch 256, relu; the only axis is `topology`). All three succeeded. AUC higher-better; log-loss and Brier lower-better. Winner bolded; best cell per metric bolded.

Arm / run id	Topology	AUC	log-loss	accuracy	Bri
T-mlp (`run-20260612-140116-41fc9-burn-deep-classifier`)	**mlp**	**0.7001**	**0.6211**	**0.6470**	**0.21**
T-emb (`run-20260612-140116-0eda4-burn-deep-classifier`)	embeddings	0.6934	0.6251	0.6436	0.21
T-wd (`run-20260612-140116-e85be-burn-deep-classifier`)	wide_deep	0.6918	0.6281	0.6407	0.21

Table 8: Phase 2 — tuning the winning `mlp` topology, a `lr`×`hidden` grid (other params held). All six succeeded, sorted by AUC; winner bolded, best cell per metric bolded. The whole grid is inside one noise band, every [128, 64] cell beats every [256, 128] cell, and the best gain over the Phase-1 winner (+0.0037) is sub-band — so the Phase-1 config is the config of record.

Run id	lr	hidden	AUC	log-loss	accuracy	Brier
`run-20260612-140211-6575a-burn-deep-classifier`	**0.0005**	**[128,64]**	**0.7038**	0.6221	0.6521	0.2172
`run-20260612-140211-21ad5-burn-deep-classifier`	0.002	[128,64]	0.6997	0.6227	**0.6532**	0.2174
`run-20260612-140211-18c9e-burn-deep-classifier`	0.001	[128,64]	0.6996	**0.6213**	0.6447	**0.2170**
`run-20260612-140211-fecb0-burn-deep-classifier`	0.0005	[256,128]	0.6966	0.6232	0.6424	0.2179
`run-20260612-140211-4c359-burn-deep-classifier`	0.002	[256,128]	0.6963	0.6245	0.6460	0.2181
`run-20260612-140211-0f4e1-burn-deep-classifier`	0.001	[256,128]	0.6957	0.6252	0.6424	0.2187

revised — a raw-one-hot Rust net reaches ∼0.70. Second, and against the optimistic reading, the representation fix does *not* reach the tree tier: at 0.7038 the net is still ∼1 band below `catboost` and ∼1.3 below `xgboost`. The gap narrowed dramatically (from the pyramid's ∼3.3 bands to ∼1) but did not close, so the convergent finding survives intact — trees still read the sparse identity one-hots a hair more effectively than even a raw-one-hot net. The topology comparison sharpens this: the plain `mlp` beating both the embedding and wide-deep variants is the *same* "the identity is already in the one-hots" lesson the tree campaigns taught, now reproduced inside the ANN family — learned per-group embeddings have little to add over 61 artist and 41 genre one-hots, and the standing lever for the net (as for the trees) is identity *resolution*: raise the vocabulary caps and the embedding topology should finally start to pay off — a prediction that the high-vocab re-run of Section 6.8 then confirms, flipping `embeddings` above `mlp` and lifting the net into the tree tier. No definition was saved here (below the ∼0.71 bar); the net is a strong, now-honest dense baseline and a natural ensemble member for its decorrelated inductive bias, not a champion. *Source: `2026-06-12-burn-deep-topology.md`.*

6.8 Confirmed: raising identity cardinality lifts the embedding net into the tree tier

Rationale. Section 6.7 closed on a sharp, testable prediction: the entity-`embeddings` topology lost to plain `mlp` *only* because the identity cardinalities (artist 61, genre 41) were too low to feed a learned embedding, so raising the vocabulary caps should flip the ranking. This campaign is the paired re-run that tests it directly. A new matrix (`ds-20260612-143014-p64-s42`) was built *identical* to the 60/40 full-64-d dataset of Section 6.7 — same 23 529 rows, same split seed 42,

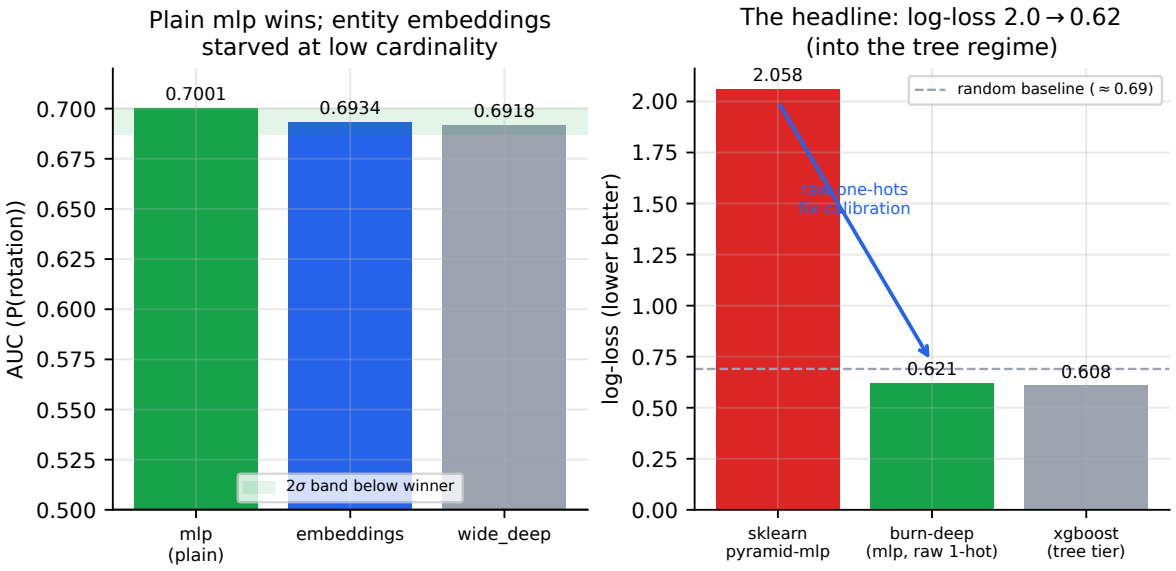


Figure 9: The representation-aware net resolves the open question of Section 6.6. *Left:* the three topologies are inside one 2σ band of each other; the plain `mlp` wins and the entity `embeddings` barely edge `wide_deep`, because at these identity cardinalities (artist 61, genre 41, album-type 4) learned embeddings are starved — they pay off at hundreds-to-thousands of categories, not dozens (a hypothesis Section 6.8 confirms by raising the caps). *Right:* the headline. Keeping the one-hots *raw* drops the log-loss from the sklearn pyramid's ≈ 2.0 (confidently miscalibrated, worse than the random baseline) to ≈ 0.62, into the tree regime — a clean confirmation that the calibration pathology was a representation artifact, not an intrinsic property of nets on this data.

byte-identical PCA64 — differing in *one* axis only: identity cardinality, with the artist cap raised 60 → 300 (50.4 % coverage) and `genre_primary` 40 → 120 (78.8 %). It is a *third* non-comparable matrix (497 columns: 64 PCA + 9 numeric + 426 one-hots — artist 301, genre 121, album-type 4 — versus the 60/40 matrix's 177), so all metrics here are fresh in-dataset references and must never be read against the 60/40 or DS-META boards. The headline question: with cardinality 5× (artist) and 3× (genre) larger, does `embeddings` now beat `mlp`? Phase 0 re-ran the two trees for an in-dataset tree tier; Phase 1 re-ran the topology bake-off (the embeddings arm now at embed_dim 16, since higher cardinality justifies a wider table); Phase 2 tuned the winning topology over embed_dim × lr.

Results. The ranking **flipped** (Table 9; Figure 10). `embeddings` (`run-20260612-143411-e8db6-burn-deep-classifier`) wins at AUC 0.7192, with `mlp` at 0.7129 (+0.0063) and `wide_deep` last at 0.7061 (+0.0131). On the 60/40 matrix `mlp` had beaten `embeddings` by +0.0067; here `embeddings` beats `mlp` by +0.0063 — a **swing of** +0.0130 in the embedding's favour, crossing the band, the qualitative reversal the starvation hypothesis predicted. Phase 0 shows both trees also moved *up* on the wider matrix: xgboost 0.7204 → 0.7299 (+0.0095, `run-20260612-143411-02fda-xgboost-classifier`, log-loss 0.6022) and catboost 0.7111 → 0.7252 (+0.0141, `run-20260612-143411-8f15e-catboost-classifier`), so high vocab is genuine signal for *every* family — it does not hand the embedding a unique advantage, it *unlocks* a topology that was previously starved. The Phase-2 embed_dim×lr tune (best embed_dim 8 / lr 5×10^{-4}, AUC 0.7228, `run-20260612-143508-3310f-burn-deep-classifier`) shows the win came from *enabling* embeddings plus a slower learning rate, not from a wider table: AUC is monotone-decreasing in embed_dim at the

slow lr (ed8 0.7228 > ed16 0.7210 > ed32 0.7200, a sub-band ∼0.0028 spread), so embed_dim is already saturated at 8. The best tuned cell beats the Phase-1 winner by only +0.0036 (sub-band), so the decision rule saved the *Phase-1* config (embeddings, embed_dim 16, lr 1×10^{-3}) as `burn-deep-embeddings-highvocab`. A determinism twin reproduced the Phase-1 run to four digits (0.71923 versus 0.71923). At 0.7192 the saved net sits −0.0060 below catboost (< 1 band) and −0.0107 below xgboost, inside the band of the second tree family — the first ANN to reach the in-dataset tree tier and clear the rotation save bar.

Table 9: Phase 1 — the `burn-deep-classifier` topology bake-off on the **high-vocab** matrix (`ds-20260612-143014-p64-s42`, held params: epochs 250, patience 20, hidden [128, 64], embed_dim 16, dropout 0.1, lr 1×10^{-3}, batch 256, relu; the only axis is `topology`). All three succeeded. The ranking *reverses* the 60/40 result of Table 7: `embeddings` now wins. AUC higher-better; log-loss and Brier lower-better. Winner bolded; best cell per metric bolded.

Topology / run id	Topology	AUC	log-loss	accuracy	B
T-emb (`run-20260612-143411-e8db6-burn-deep-classifier`)	**embeddings**	**0.7192**	**0.6087**	**0.6694**	**0.**
T-mlp (`run-20260612-143411-01d7b-burn-deep-classifier`)	mlp	0.7129	0.6166	0.6617	0
T-wd (`run-20260612-143411-7d117-burn-deep-classifier`)	wide_deep	0.7061	0.6256	0.6526	0

Table 10: Phase 2 — tuning the winning `embeddings` topology, an embed_dim×`lr` grid (hidden [128, 64], other params held), sorted by AUC; winner bolded, best cell per metric bolded. AUC is monotone-decreasing in embed_dim at lr 5×10^{-4} (the table is already saturated at width 8), and lr 5×10^{-4} beats lr 1×10^{-3} at every width. The best gain over the Phase-1 winner (+0.0036) is sub-band, so the Phase-1 config is the config of record.

embed_dim	lr	AUC	log-loss	accuracy	Brier
8	**0.0005**	**0.7228**	0.6105	0.6645	0.2114
16	0.0005	0.7210	**0.6088**	**0.6711**	**0.2108**
32	0.0005	0.7200	0.6100	0.6657	0.2114
8	0.001	0.7193	0.6092	0.6643	0.2112
16	0.001	0.7192	0.6087	0.6666	0.2110
32	0.001	0.7183	0.6097	0.6668	0.2114

Interpretation. This resolves Section 6.7's open question *confirmed*: the embedding topology's earlier loss was a starvation artifact of low cardinality, not an intrinsic limit, and raising the caps flips the ranking exactly as predicted. Two honest caveats keep the claim calibrated. First, the single-split emb-vs-mlp margin (+0.0063) is itself sub-band, so the *magnitude* of the reversal needs a 3-split confirm; what is decisive is the *direction* — the whole ranking reordered, with the embedding arm moving from the bottom of the 60/40 ranking to the top of this one. Second, high vocab helps *everyone*: the trees gained as much as (catboost) or more than the embedding arm, so the correct reading is "higher resolution is better signal for every family, and it specifically unlocks the embedding topology", not "embeddings now win outright" — `xgboost` at 0.7299 is still the in-dataset leader, ∼0.0107 (∼1 band) above the saved net. But the qualitative arc that ran through Section 6 now closes cleanly: a dense net that consumes raw one-hots is calibrated in the tree regime (Section 6.7), and given *enough* identity resolution its embedding topology reaches the tree tier — the standing "identity resolution is the lever" lever, demonstrated. Logloss (∼0.61) and Brier (∼0.21)

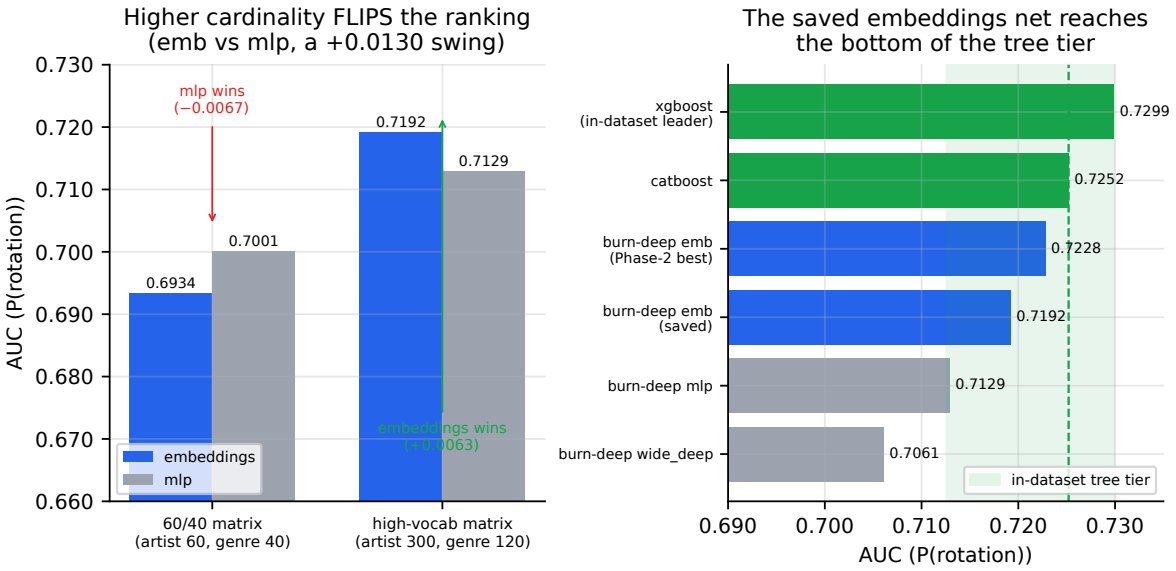


Figure 10: Identity resolution closes the ANN-pitfalls thread. *Left:* the paired `embeddings`-versus-`mlp` comparison across the two cardinalities. On the 60/40 matrix `mlp` wins by +0.0067; on the high-vocab matrix `embeddings` wins by +0.0063 — a +0.0130 swing that reverses the ranking and confirms the starvation diagnosis: with artist cardinality 5× and genre 3× larger the embedding table finally has enough distinct identities to learn a useful low-rank geometry instead of collapsing near-constant. *Right:* the five-family in-dataset AUC ranking on the high-vocab matrix. The saved `embeddings` net (0.7192) reaches the bottom of the tree tier (shaded: from a band below `catboost` up to `xgboost`) — the first ANN to do so — while `xgboost` (0.7299) remains the in-dataset leader.

stay in the trees' calibrated regime throughout, so there is no calibration penalty for the win. One data-hygiene note: the high-vocab manifest's redundancy report flags 21 near-perfect genre↔artist one-hot collinearities (e.g. `genre_primary=australian psych` ≡ `artist=Tame Impala`, correlation 1.000; 19 more ≥ 0.957), an artifact of admitting rare genres carried by a single newly-admitted artist; the trees and the net's regularizers absorb it with no observed run pathology, but it is worth knowing before reading individual one-hot importances on this matrix. The saved `burn-deep-embeddings-highvocab` is flagged for best-model-selector as the first tree-tier ANN and a natural family-diversity candidate. *Source:* `2026-06-12-high-vocab-topology.md`.

7 Conclusions

Seven Jun-2026 campaigns on the pivoted rotation target converge on a single story. (1) Across two feature campaigns and three model-family scans (two corpora), every dense modality and every dense distance/margin/dense-net family loses to categorical identity one-hots read by gradient-boosted trees — the song-AE latent dilutes (Section 5.1), kernel SVM is worse-than-mean (Section 5.2), and on the new full-64-d corpus the two tree families lead while the dense pyramid-MLP is last (Section 5.4). (2) The choice of metric is load-bearing on a binary target: MAE is degenerate (a two-level ϵ-tube predictor posts a deceptively low MAE while being worse than the mean), and a grid-by-MAE blend even mis-selects such a predictor; the honest classification metrics (AUC, log-loss, Brier) restore a fair comparison (Section 5.3). (3) The campaigns delivered two measured noise bands — $2\sigma = 0.0086$ MAE and $2\sigma = 0.0126$ AUC — and the project's first registered, pro-

motable rotation champion, `xgboost-classifier-rotation` at AUC 0.7429 on DS-META, plus a saved new-corpus xgboost definition at AUC 0.7204. (4) The dense neural net was diagnosed in full (Section 6): an architecture scan refuted the budget hypothesis (every geometry self-stops at 73–100 iterations and the converged net reproduces the under-trained AUC to all digits), so the binding constraint is the raw one-hot *representation*, not optimization — and the net's worse-than-random log-loss beside a genuine AUC marks it as confidently miscalibrated, not merely weak. (5) That diagnosis was then tested directly: the Rust/`burn` `burn-deep-classifier`, which feeds the identity one-hots *raw*, repaired the calibration completely (log-loss 2.0 $\rightarrow\sim$ 0.62, into the tree regime) and lifted AUC $\sim$2 bands to 0.7038 (Section 6.7) — confirming the pathology was a representation artifact, not an intrinsic property of nets. But the gap to the trees only *narrowed* (from $\sim$3.3 bands to $\sim$1), it did not close: the net still lands below `catboost` and `xgboost`, the plain `mlp` beat both the embedding and wide-deep topologies (entity embeddings are starved at the corpus's low identity cardinality), and nothing cleared the $\sim$0.71 save bar. (6) That standing lever was then pulled. A paired re-run on a *high-vocab* matrix — identical to the 60/40 dataset but with the artist cap raised 60 $\rightarrow$ 300 and genre 40 $\rightarrow$ 120 — *flipped* the topology ranking (Section 6.8): `embeddings`, starved and last on the 60/40 matrix, now *beats* `mlp` (a +0.0130 swing that reverses the prior result and confirms the starvation hypothesis directionally), and the saved net reaches the in-dataset tree tier at AUC 0.7192 (within $\sim$1 band of `catboost`) — the first ANN to clear the rotation save bar, saved as `burn-deep-embeddings-highvocab`. The convergent finding still stands at the top (`xgboost` leads every matrix, and higher resolution lifts *every* family, not just the embedding), but the ANN-pitfalls thread is now closed: a representation-aware net that consumes raw one-hots is calibrated in the tree regime, and given enough identity resolution its embedding topology reaches the tree tier — identity *resolution* was the lever all along.

8 Source reports

This synthesis consolidates seven campaign reports from `experiments/`:

- `2026-06-09-ae-xgboost-blend.md` — autoencoder-latent + XGBoost blend on rotation (Section 5.1).
- `2026-06-09-svm-kernel-rotation.md` — kernel SVM scan and conditional blend; first measured MAE noise band (Section 5.2).
- `2026-06-09-rotation-classification.md` — AUC bake-off, first measured AUC noise band, first registered rotation champion (Section 5.3).
- `2026-06-12-classifier-bakeoff-p64.md` — four-family classifier bake-off and xgboost depth/lr sweep on the new full-64-d corpus (Sections 5.4 and 6).
- `2026-06-12-pyramid-mlp-arch-scan.md` — six-geometry pyramid-MLP architecture scan, refuting the budget hypothesis (Sections 5.5 and 6).
- `2026-06-12-burn-deep-topology.md` — the Rust/`burn` `burn-deep-classifier` topology bake-off; resolves the representation-aware-net question, fixing calibration but not the AUC gap (Section 6.7).
- `2026-06-12-high-vocab-topology.md` — the paired topology re-run on a higher-identity-cardinality matrix; flips `embeddings`>`mlp` and lifts the saved net into the in-dataset tree tier, the first ANN to clear the rotation save bar (Section 6.8).